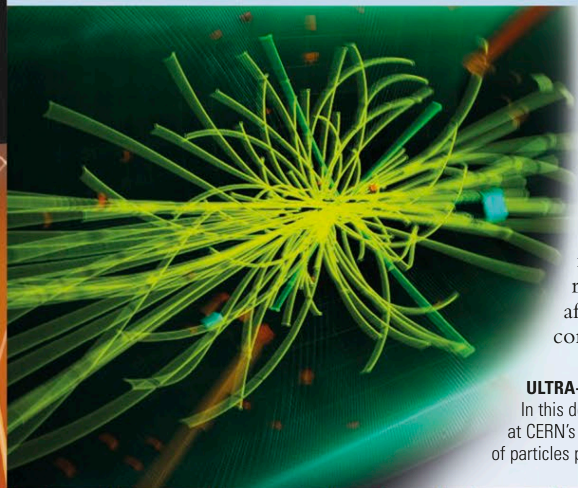
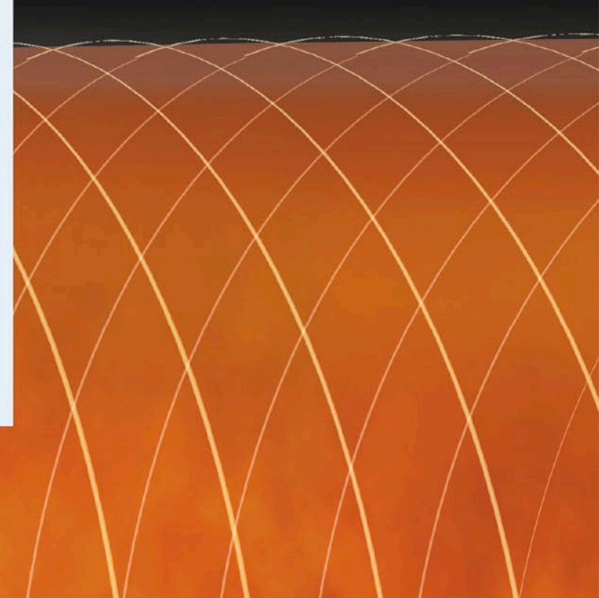
EXPLORING SPACE

RECREATING THE EARLY UNIVERSE



At the European Center for Nuclear Research, also known as CERN, particle physicists are unraveling the finer details of the early universe by smashing particles together in particle accelerators and searching for traces of other elementary particles. In doing so, they explore the constituents of matter and the forces that control their interactions. CERN scientists have even recreated conditions like those shortly after the Big Bang by creating plasmas containing free quarks and gluons.

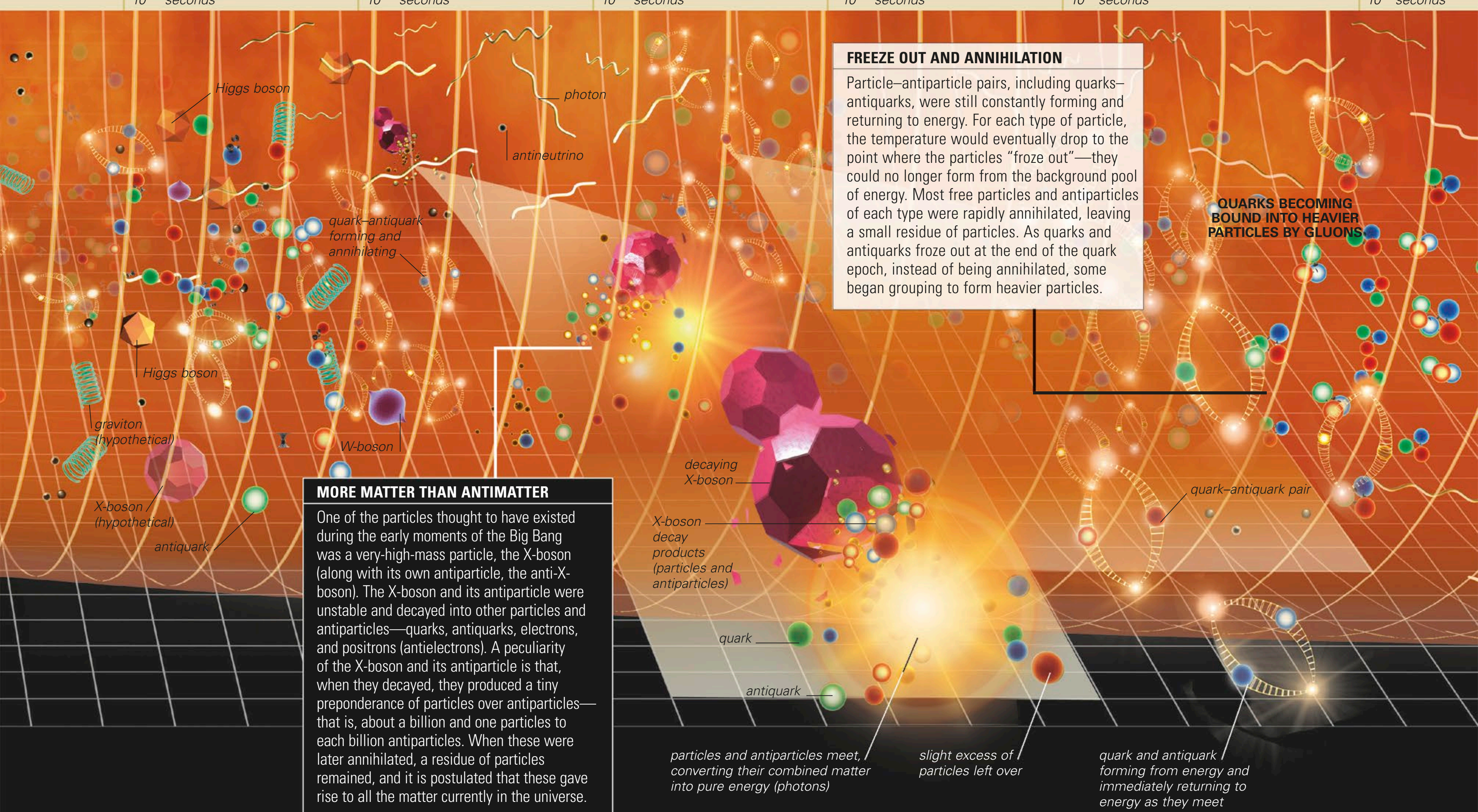
ULTRA-HIGH-ENERGY PROTON COLLISION
In this depiction of activity within one of the main experiments at CERN's Large Hadron Collider, the yellow lines are the paths of particles produced by a proton-proton collision.



	10^6m (620 miles/1,000 km)	10^9m (620,000 miles/1 million km)	10^{12}m (620 million miles/1 billion km)
billion trillion °C)	10^{21}K (1.8 billion trillion °F/1 billion trillion °C)	10^{18}K (1.8 million trillion °F/1 million trillion °C)	10^{12}K (1.8 trillion °F/1 trillion °C)

QUARK EPOCH
At the start of this epoch, the electroweak force separated into the electromagnetic and weak forces, and physical laws became as they are today. By its end, temperatures had dropped to a point where gluons could bind quarks together.

1 zeptosecond 10^{-21} seconds | *1 attosecond 10^{-18} seconds* | *1 femtosecond 10^{-15} seconds* | *1 picosecond 10^{-12} seconds* | *1 nanosecond 10^{-9} seconds* | *1 microsecond 10^{-6} seconds*



FREEZE OUT AND ANNIHILATION
Particle–antiparticle pairs, including quarks–antiquarks, were still constantly forming and returning to energy. For each type of particle, the temperature would eventually drop to the point where the particles "froze out"—they could no longer form from the background pool of energy. Most free particles and antiparticles of each type were rapidly annihilated, leaving a small residue of particles. As quarks and antiquarks froze out at the end of the quark epoch, instead of being annihilated, some began grouping to form heavier particles.

MORE MATTER THAN ANTIMATTER
One of the particles thought to have existed during the early moments of the Big Bang was a very-high-mass particle, the X-boson (along with its own antiparticle, the anti-X-boson). The X-boson and its antiparticle were unstable and decayed into other particles and antiparticles—quarks, antiquarks, electrons, and positrons (antielectrons). A peculiarity of the X-boson and its antiparticle is that, when they decayed, they produced a tiny preponderance of particles over antiparticles—that is, about a billion and one particles to each billion antiparticles. When these were later annihilated, a residue of particles remained, and it is postulated that these gave rise to all the matter currently in the universe.